2025 Immcantation Users Group Meeting

Organized by:
Kleinstein Lab
Yale University

Immcantation Users

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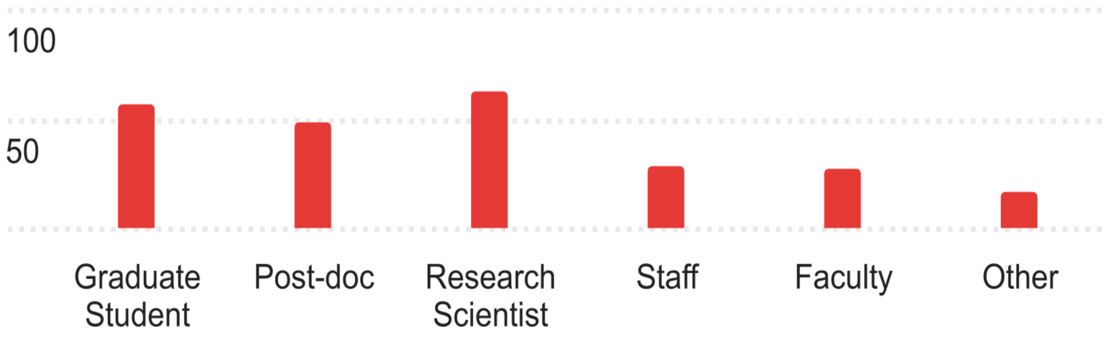


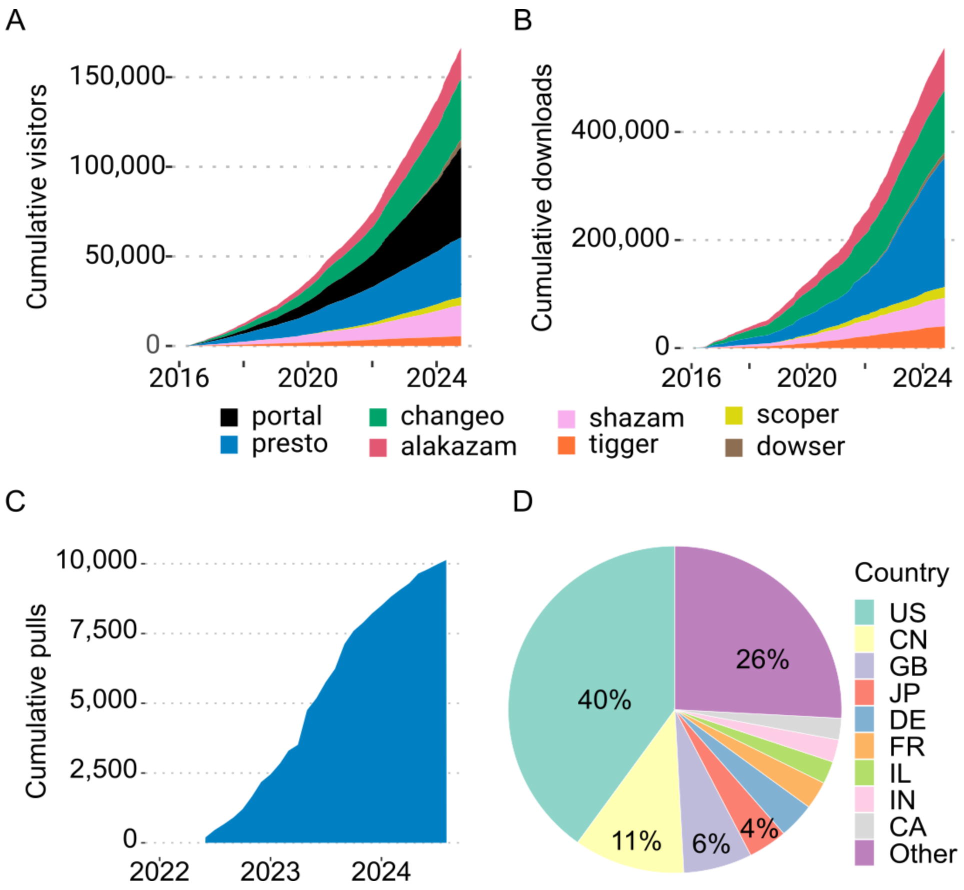
90% academic

"We mainly focus on TCRs, but would still like to join the meeting since many aspects are similar in TCR/BCR analysis"

"Interested in epitope-specific BCR sequence discovery, so hoping to see some talks on that."

"I've been dabbling/fighting with a variety of these sorts of tools, happy to have the opportunity to listen in and learn more."

Immcantation users



Start-to-finish analytical ecosystem

Raw sequences	V(D)J Alignment	Annotated repertoire	Biological interpretation
pRESTO Read Assembly Quality Control UMI processing	**IMGT** **IgBLAST**	**Change-O** Alignment Standardization Clonal Clustering	**Alakazam** Diversity Analysis V(D)J Gene Usage Physicochemical Properties
		SCOPer Clonal Clustering	**SHazaM** Mutation Profiling Selection Quantification
		TIgGER Genotyping Polymorphism Detection	**Dowser** B cell lineage trees Migration & differentiation Evolution over time

AGPLv3 Free Software
Free as in Freedom

R python docker
Bitbucket GitHub

immcantation.org | immcantation@googlegroups.com | http://airr-community.org

AIRR-C sw-tools v1 compliant

Goals for 2025

Continue to expand the framework

- Tutorials
- Single cell
- AIRRflow
  - Nextflow best-practice workflow
  - Bulk and single cell
  - https://nf-co.re/airrflow
  - Reports
  - Performance
  - Tutorials

Organizational changes

- Multi-lab/institutional contributions
  - Hoehn Lab
    - Add functions to reconstruct full-length germline sequences, as well as continue improving methods for building trees with single cell and bulk BCR data. Dowser, IgPhyML.
  - Yaari Lab
    - Lead germline and allele inference efforts. TIgGER, PIgLET and RAbHIT.
- Move to GitHub
  - Facilitate contributions. Guidelines.
  - Improve testing. Automation.

Resources, support, contact

Documentation and tutorials
https://immcantation.org

Questions and issues
immcantation@googlegroups.com

Source code, bug reports, feature requests, issues
https://github.com/immcantation/<tool-name>

Docker container
https://hub.docker.com/r/immcantation/suite

News
https://groups.google.com/g/immcantation-news

Kleinstein Lab
https://medicine.yale.edu/lab/kleinstein

steven.kleinstein@yale.edu

Inquire about openings. We are always looking to welcome postdocs and research staff.

steven-kleinstein

@skleinstein.bsky.social

Funding
R01 AI104739
U19 AI089992
T15 LM07056
BSF 2013395

Current Immcantation Team

Kleinstein Lab (Yale)

Steven Kleinstein, PhD
Anthony N Brady Professor of Pathology

Edel Aron
PhD Candidate

Paris Filippidis, MD
Postdoctoral Fellow

Gisela Gabernet, PhD
Associate Research Scientist

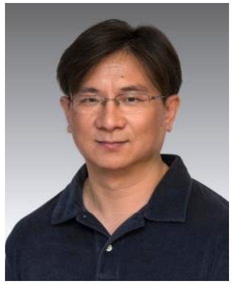
Cole Jensen, MPH
PhD Candidate

Edward Lee, MD, PhD
Clinical Fellow

Noah Yann Lee
PhD Candidate

Huimin Lyu
Bioinformatics Programmer

Susanna Marquez
Bioinformatician



Hailong Meng, PhD
Senior Bioinformatics Scientist

Burhan Sabuwala
PhD Student

Yaari Lab (Yale)

Gur Yaari, PhD
Associate Professor

Ayelet Peres, PhD
Postdoctoral Fellow

Hoehn Lab (Dartmouth)

Kenneth Hoehn, DPhil
Assistant Professor

Jessie Fielding
Research Scientist

Hunter Melton, PhD
Postdoctoral Research Fellow

Sherry Wu
PhD Student

Thanks everyone for coming!!

- We will send out speaker answers to unanswered questions as well as a copy of these slides.
- Please fill out the post-event survey (it will be emailed out to all registered people)! We greatly appreciate your feedback.
- The videos (for everyone who agreed to be recorded) will be available online on the Immcantation YouTube.

https://immcantation.readthedocs.io

immcantation@googlegroups.com

AIRR Community: Virtual Meeting III

Save the date: The Zooming into the AIRR Community III virtual meeting will take place on **May 21st & 22nd, 2025** (8:00 PST, 17:00 CEST - 12:00 PST, 21:00 CEST)

This virtual event will consist of two half-day sessions that will include the celebration of the AIRR Community 10th anniversary, a brief General Assembly session, scientific talks, and round table "hot topic" breakout discussions.

AIRR Community

antibodysociety.org/the-airr-community/meetings/

Immunoinformatics Special Issue

Perspective of Machine Learning and Germline Variation Research In Adaptive Immune Receptor Repertoires

Submission deadline: **15 February 2025**

Since its inception, the Adaptive Immune Receptor Repertoire Community (AIRR-C) has fostered collaborative and structuring works across diverse disciplines, spearheading initiatives to standardize AIRR-sequencing data practices, develop shared resources, and promote innovative research.

This coming year (2025), the AIRR-C will be celebrating its 10th anniversary. To commemorate this historical milestone for our organization, we are compiling a collection of articles that highlight the culmination of key AIRR-C initiatives from the past decade, as well as emerging themes in the AIRR research domain. This special issue will center around reports from our most recent meeting, "AIRR-C Meeting VII: Learnings and Perspectives", from which we will seek related submissions from AIRR-C members and the broader research community. A special emphasis will be placed on articles that showcase (1) the importance of immune receptor gene germline variation and its impact on immune function, and (2) the development and application of machine learning approaches for untangling complexity in AIRR datasets in the context of health and disease.